

Project 17: Explainable Food Recommendation, Explanation Quality Analysis & Enhancement. (Proposed by Amir Mollazadeh)

Goal

Analyze and improve the quality of natural language explanations generated by a hybrid food recommendation system. Your task is evaluating whether explanations are factually accurate, clear, transparent in internal reasoning, and free from demographic bias. You will perform NLP-based analysis on existing explanations, then generate and compare explanations from multiple LLMs to see which produces higher-quality outputs.

Dataset

The XFoodRec project has already generated:

- **personas.json**: ~40 synthetic user profiles with dietary goals, allergies, preferences, demographics
- **recommendations.json**: Generated recommendations with explanations for each persona (~15 recipes per persona = ~600 total recommendation sets). [Generated by GPT-4o-mini]
- **recipes.parquet**: Source recipe data (nutrition, ingredients, tags,...)

[XFoodRec GitHub: <https://github.com/Amir-Mol/XFoodRec>]

Specifications

1. Data Exploration & Preparation

- Load and inspect **personas.json**, **recommendations.json**, and **recipes.parquet**
- Extract all explanations and compute basic statistics:
 - Length (words/characters)
 - Distribution of explanation types (ingredient-, nutrition-, taste-focused)
 - Coverage of user preferences vs. health justification

2. Entity Extraction, Fact Verification & Hallucination Detection

- Extract three types of claims from each explanation: 1. Ingredient mentions (e.g., "garlic", "salmon"), 2. Nutrient claims, qualitative or numeric (e.g., "high in protein", "15g protein"), 3. Diet/health-label claims (e.g., "gluten-free", "diabetes-friendly", "low sodium")
- Compare and verify extracted claims with the actual data:
 - Ingredients → recipe's ingredient list
 - Nutrient claims → recipe's nutrition fields
 - Diet/health-label claims → both the recipe's ingredients and the persona's restrictions/allergies/conditions
- Define hallucination types:
 - Ingredient: mentions an ingredient not in the recipe
 - Nutrition: claimed value significantly differs from actual data (numeric) or contradicts the documented threshold (qualitative)
 - Diet/allergen: claims a diet/health label or compatibility with a persona restriction that the recipe's ingredients contradict (or vice versa)
- Validate the verifier itself: Manually label a sample of explanations as hallucinated/not.
- Report:
 - Hallucination rate (% of explanations with ≥ 1 false claim), broken down by type
 - Patterns (do longer explanations or certain persona types hallucinate more?)

3. Transparency & Reasoning Analysis

The XFoodRec prompt requires explanations to explicitly cite the user factors (goal, preference) that drove the recommendation

- Classify each explanation by transparency level:
 - High: explicitly mentions ≥ 1 user factor, e.g., "matches your muscle gain goal"
 - Medium: implies user relevance but not explicit
 - Low: generic, no user context
- - Report the distribution of levels and compare across persona types.
Note: You can classify with rule/lexicon matching (a list of factor keywords + user-directed cue words such as "your"/"goal"/"needs", excluding words that only appear in the recipe title)

4. Bias & Fairness Analysis

- Investigate whether explanation quality varies by persona characteristics:
 - Dietary goal
 - Health status
 - Dietary restriction
 - Demographics (age, gender)
- Metrics:
 - Average explanation length
 - Transparency score
 - Hallucination rate
- Apply statistical tests (t-tests, ANOVA) on persona-level values with multiple-testing correction, and frame results as exploratory.
Note: The explanations are nested in 40 personas, so it is good to aggregate metrics per persona before testing.

7. Multi-Model Explanation Generation

- Use at least 2 different LLMs (you can use Lehmus AI platform or other services available for University of Oulu students) to generate explanations for the exact same persona + recipe + rank triples already in **recommendations.json**
Note: choose LLMs that differ meaningfully in size/architecture for a more informative comparison.
- Run each model's explanations through the Task 1-4 pipeline for comparison against the original GPT-4o-mini explanations.

9. Evaluation & Comparison

- Compare original (GPT-4o) explanations vs. new models output
- Metrics:
 - Factual accuracy (% of verified claims)
 - Transparency score
 - Hallucination rate
 - Average length
- Report a model-comparison table and discuss which model(s) produced better explanations and why .

Deliverables

- Code/notebook implementing all specifications above.
- The raw explanation outputs collected from each additional model.

- A written report covering all analyses, with tables/plots, the model comparison table, the model comparison table, and the hallucination and bias findings.

Project 18: Multilingual Clinical Text Summarization (proposed by Aleka Ayalew)

1. Objective

The objective of this project is to investigate how diverse NLP methods generate concise summaries from clinical text across multiple languages. You will implement and compare classical extractive summarization, multilingual transformer models, and instruction-tuned LLMs. The project evaluates how summarization quality varies across native benchmark languages (English and Spanish or French) and a translated test set (Finnish), examining document lengths, compression rates, and model hallucinations.

2. Dataset Specification

You will use the open-access MultiClinSum benchmark:

- **Dataset Link:** <https://zenodo.org/records/17341582>
- (Alternative : <https://zenodo.org/records/15459174>)
- **Content:** Clinical case reports paired with human-written reference summaries in English (`en`), Spanish (`es`), French (`fr`), and Portuguese (`pt`).
- **Language Scope:** You will evaluate on three languages:
 1. English (`en`) directly from the benchmark.
 2. One Romance language directly from the benchmark: Spanish (`es`) or French (`fr`).
 3. Finnish (`fi`) translated set: Generated by translating the English clinical cases and reference summaries into Finnish (using `Helsinki-NLP/opus-mt-en-fi` or an open instruction LLM), clearly labeled as translated data.

3. Summary Output Length and Parameter Specifications

- **Extractive Summaries:** Recommended target of 20% to 25% of source sentences (or top 4–6 sentences). You can adjust this target as you want to get better results.
- **Transformer Generation Parameters:**
 - Maximum token limit: `max_new_tokens = 512`.
 - Recommended baseline starting values: `num_beams = 4`, `length_penalty = 1.0` to `1.2`, `no_repeat_ngram_size = 3`.
 - **Parameter Adjustment:** You are not restricted to fixed values. You can adjust and tune parameters as you want (such as beam search, length penalty, repetition penalty, or sampling settings) to get the best possible summary quality and prevent text from repeating or cutting off. Keep chosen settings consistent when comparing across languages.